a
KEGG pathway data
23
pathway maps
3,207
typed relations
860
genes
2,176
unique gene pairs
act.
inh.
phos.
b
Hidden shortcuts
19.6%
of pairs carry two labels
41%
of random-split test rows
reuse a training pair
MAPK 1
MAPK 3
432
gene-name families
share informative stems
0.842
ceiling on single-label accuracy
c
Leakage-aware splits
stricter
Pair-disjoint
unseen pairs, seen genes
10 folds
Gene-disjoint
unseen genes
10 folds
Family-disjoint
unseen name families
10 folds
Pathway held out
unseen pathways
23 folds
d
AI and language models
gene pair
M A P K 1
?
E L K 1
Symbol models
LR-Lex
CharCNN
Language models
BioBERT v1.1, v1.0
SciBERT, SPECTER
Graph models
Lookup
LR-Graph
GBDT-Graph
GraphSAGE GNN
act.
inh.
phos.
multi-label
probabilities
e
Stress tests
Scrambled names
original
scrambled
1
0.5
P G F Pw
Swapped roles
original
swapped
1
0.5
P G F Pw
Calibration (ECE)
LR-Lex
GBDT-Graph
0.4
0.2
0
P G F Pw
Conformal coverage
--- 90% target
1
0.5
0
91
85
73
58
P G F Pw
f
Key finding
AUROC
1
0.75
0.5
phos. 0.87
act. 0.62
inh. 0.56
near chance
Pair
Gene
Family
Pathway